- **Unified data asset governance**: VS integrates with Unity Catalog; Unity Catalog handles data governance and access control lists. To prevent productional data leakage, you can manage access to the Databricks VS API and the underlying databases with Unity Catalog.
- **Model Serving integration**: Model Serving automates querying of the model serving endpoint for embedding generation without any overhead from users.

Flexibility in embedding model support

One of VS's major strengths is its support for any of the embedding models of your choice. VS supports hosted or fully managed embeddings. Hosted embeddings are self-managed. You create the embeddings and save them on a Delta table. For fully managed embeddings, the prepared text is saved in a Delta table, and embeddings are created by Databricks Model Serving. Model Serving will convert your incoming data into embeddings with the model of your choice. Databricks VS can support any model through the Databricks Model Serving endpoints, the Foundation Model API (as mentioned in *Chapter 3*), or external models. External models include SaaS models, for example, OpenAI's ChatGPT and Anthropic's PaLM. You can connect external models through the Databricks unified model serving gateway. See *External models in Databricks Model Serving* in *Further reading* for more information.

Setting up a vector search

As mentioned, VS is a serverless product. Hence, we require a real-time connection with our chatbot application to the relevant content stored in the vector database. We'll cover this again in the *Applying our learning*, *Project – RAG chatbot* section, but if you're ready to set up your own endpoint, you just need a few lines of code, as shown in *Figure 4.15*.

```
from databricks.vector_search.client import VectorSearchClient

vsc = VectorSearchClient()
vsc.create_endpoint(name="my_vector_search_endpoint", endpoint_type="STANDARD")
```

Figure 4.15 – Creating a Databricks VS endpoint

Once you create an endpoint, you can host multiple vector search indices under one endpoint. The endpoint will scale according to the demand. The code to host your first index is presented in the *Apply our learning* section).

There are limitations on the number of endpoints you can create indices per endpoint, and the embedding dimensions. Please review the documentation linked in *Further reading*, as Databricks may remove or update limits as the product evolves.

Technology moves fast, especially in the world of generative AI. Databricks VS was built even as we wrote this book, and we expect it to continue evolving to search not only through text but also images and audio with a more robust hybrid search engine in the future.